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Unit – 4 : Session – 7 : SLO – 1 | Binary Search Trees — Insertion Operation

■ I. Fill in the Blanks

- In BST insertion, first compare the new value with the ROOT (current node).
- If the new value to be inserted is smaller, go to the LEFT child.
- If the new value to be inserted is larger, go to the RIGHT child.
- To insert a new element in BST; continue until an appropriate LEAF (NULL) position is found.
- After finding appropriate position the new node is inserted at the LEAF (NULL link) location.

■ II. Algorithm to Insert a Value in a BST

- Insert (TREE, VAL)
- Step 1: IF TREE = NULL
- Allocate memory for TREE
- SET TREE -> DATA = VAL
- SET TREE -> LEFT = TREE -> RIGHT = NULL
- ELSE
- IF VAL < TREE -> DATA
- Insert(TREE -> LEFT, VAL)
- ELSE
- Insert(TREE -> RIGHT, VAL)
- [END OF IF]
- [END OF IF]
- Step 2: END